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Hand preferences and differences in extractive foraging in seven capuchin monkey species

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Tool use and extractive foraging could be drivers for right hand use preference. The robust capuchins, *Sapajus*, are more specialized for destructive and extractive foraging than the gracile capuchins, *Cebus*. Thus, we predicted them to show right-hand preference and higher rates of extractive foraging when compared to the gracile capuchins. We hypothesized that capuchins that evolved in dry habitats are predisposed to show higher rates of extractive foraging and right-hand preference. We employed the tube task (152 individuals) and an extractive foraging task (212 individuals) in seven species of capuchins in captive settings. Data on hand preference and extractive foraging were collected using focal and scan sampling, respectively. Contrary to our hypothesis, *S. libidinosus* showed significant left hand preference, while *S. robustus* and *S. flavius* showed a right hand preference. The species *S. xanthosternos*, *S. flavius*, and *S. robustus* engaged in extractive foraging significantly more often than *S. apella*, *S. libidinosus*, and *Cebus albifrons*. As expected, the gracile capuchins showed significant higher left-hand bias when compared to the robust capuchins. The findings from this study is the first indication of left-hand population level laterality in *S. libidinosus*, a species that evolved in an environment selecting for increased tool use. Although tool use has been associated with population level right handedness, or a strong trend to use the right hand, our results suggest the link between tool use and right-hand preference is simplistic, at least in *Sapajus*. The ultimate explanation for the differences in hand preference across species might be due to differences in use of key food resources. We suggest that capuchin monkeys from the Atlantic forest evolved under a stronger selective pressure for high reliance on bimanual feeding and extractive foraging, which requires more sequential actions and planning, than those species from the Amazonia forest and Caatinga/Cerrado dry forest.

KEYWORDS
bimanual feeding, handedness, manual lateralization, sex differences, tool use

1 | INTRODUCTION

Lateralization of behavior and functions has been found to be widespread both in vertebrates and invertebrates. This lateralization reflects functional asymmetries of the brain that probably evolved to increase neural processing capacity, avoiding duplication of functions in the two hemispheres, and it also provides selective advantages (Rogers, Vallortigara, & Andrew, 2013; Wiper, 2017). The preeminent group level preference for the right upper limb seems to be rare in tetrapods, except for humans (Cavanagh, Berbesque, Wood, & Marlowe, 2016; Marchant & McGrew, 2013; Rogers et al., 2013; Wiper, 2017).

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In the last decades, studies evaluating hand preference in non-human primates have increased greatly, both in captive and wild populations, and various authors suggested that a right hand bias at the population level does occur in non-human primates (Hopkins et al., 2011; Humle & Matsuzawa, 2009; Llorente et al., 2011; Lonsdorf & Hopkins, 2005; Molesti, Vauclair, & Meguerditchian, 2016; Neufuss, Humle, Cremaschi, & Kivell, 2016; Poindexter, Reinhardt, Nijman, & Nekaris, in press; Regaiolli, Spiezio, & Hopkins, 2016, 2018; Spinozzi, Castorina, & Truppa, 1998; Tabiowo & Forrester, 2013), but this right hand bias is not as strong and stable across different tasks as in humans (Cashmore, Uomini, & Chapelain, 2008; Marchant & McGrew, 2013). Other studies, however, with the same and different species report that lateralization in hand preference occurs only at an individual level (Brand et al., 2017; Leca, Gunst, & Huffman, 2010; Lilak & Phillips, 2008; Marchant & McGrew, 2007; Meunier & Vauclair, 2007; Papademetriou, Sheu, & Michel, 2005; Westergaard & Suomi, 1996a).

The conflicting results of studies evaluating hand preference prompted criticisms based on methodological and theoretical issues (e.g., Cashmore et al., 2008; Marchant & McGrew, 2013). Differences in experimental protocols, large variability in tasks used in evaluating laterality, and postural constraints (Braccini, Lambeth, Schapiro, & Fitch, 2010; Cashmore et al., 2008; Westergaard, Kuhn, & Suomi, 1998) could affect hand preference in addition to species differences and sample size issues (Cashmore et al., 2008; Hopkins, 2006; Hopkins et al., 2011; McGrew & Marchant, 1999).

In the case of methodological issues, the objection seems to boil down to the type of tasks used for measuring hand preference. In general, tasks that do not require cognitive and motor demanding actions, such as the commonly-used manual reaching for food, hardly evoke hand preference in non-human primates (Fagot & Vauclair, 1991; Papademetriou et al., 2005; Schnoell, Huebner, Kappeler, & Fitchel, 2014). According to the task complexity hypothesis, proposed by Fagot & Vauclair (1991), high-level tasks, that is, those needing cognitively demanding requirements, tend to elicit stronger hand preference in non-human primates, although this hypothesis does not make predictions about direction of hand preference. Indeed, when the tasks being performed are skillful, requiring fine motor control, and cognitive abilities (e.g., activities requiring complementary use of hands, extractive foraging, and tool use) increased laterality and hand preference emerges (e.g., Bardo, Pouydebat, & Meunier, 2015; Fagot & Vauclair, 1991; Hopkins, 1995; Hopkins, Russell, & Cantalupo, 2007; MacNeilage, 2007; Marchant & McGrew, 2013). According to some authors the consequences of the increasing demand for higher motor skill associated with tool use could lead to right handedness (Frost, 1980; Hopkins et al., 2017; MacNeilage, 2007; Westergaard & Suomi, 1996b). Yet, conflicting results occur even when these cognitively demanding tasks are used (Schweitzer, Bec, & Blois-Heulin, 2007). For example, some studies show a group level left-hand bias with probing tools in wild chimpanzees (Bogart et al., 2012; Lonsdorf & Hopkins, 2005), while other studies indicate right-handedness when nut cracking with tools (Lonsdorf & Hopkins, 2005). Similar results are reported for capuchin monkeys in captivity (Anderson, Degiorgio, Lamarque, & Fagot 1996; Phillips & Thompson, 2013; Westergaard & Suomi, 1996b). Nonetheless, the contrasting results can be explained in terms of differences in the cognitive, sensory, and motor demands of these tool-use activities that are specific to one cerebral hemisphere (Cochet & Byrne, 2013; Fagot & Vauclair, 1991; Lonsdorf & Hopkins, 2005; MacNeilage, 2007).

The inconsistencies in different studies, particularly due to the varied tasks used to measure hand preference, hinder comparisons, and make it difficult to interpret findings and to formulate any general conclusion. There has been, however, an incipient standardizing in tasks used to assess hand preferences in non-human primates, particularly using bimanual coordinated tasks, that are considered a better indicator of hand and brain laterality in non-human primates than less cognitively demanding tasks (e.g., Hopkins, 2013; Hopkins & Cantalupo, 2004; Meguerditchian, Vauclair, & Hopkins 2013). In recent years, the tube task (Hopkins, 1995) has stood out as the preferred choice for evaluating laterality in numerous studies (e.g., Fan et al., 2016; Morino, Uchikoshi, Bercovitch, Hopkins, & Matsuzawa, 2017; Regaiolli, Spiezio, & Hopkins, 2018; Zhao, Hopkins & Li, 2012). The tube task consists of a tube of PVC, with variable dimensions depending on the studied species, that requires coordinated bimanual actions, that is, the primate holds the tube with one hand and uses a finger of the opposite hand, to retrieve a food reward from inside the tube. The tube task is considered a reliable and appropriate task for investigating manual asymmetries that allows consistent comparisons to be made among non-human primates (Hopkins et al., 2015; Maille, Belboc'h, Rossard, Bec, & Blois-Heulin, 2013; Regaiolli et al., 2016). Moreover, in non-human primates the tube task has been shown to elicit hand preference that is associated with neuroanatomical asymmetry within the primary motor cortex in chimpanzees and capuchin monkeys (Hopkins & Cantalupo, 2004; Phillips & Sherwood, 2005).

Several studies, using the tube task, indicate differences in hand preference across species. Some species, such as baboons and chimpanzees, show population level consistency in choosing the right hand to retrieve the food reward (Hopkins et al., 2011; Molesti et al., 2016), while other species, such as snub-nosed monkeys and siamangs (Morino, 2011; Morino et al., 2017; Zhao et al., 2012) are left handed. The ultimate explanations for population level hand preference, suggest factors such as random genetic mutation, evolutionary stable strategy, postural demands, and increased complexity of foraging that requires skilled motor control, as the evolutionary drivers (Frost, 1980; MacNeilage, Studdert-Kennedy, & Lindblom, 1987; Rogers et al., 2013).

One relatively accepted and supported explanation that could drive right hand preference at the population level in non-human primates is the Postural Origin Hypothesis put forward by MacNeilage et al. (1987). The explanatory structure of this hypothesis combines variables like arboreality, postural changes, visually guided manual predation, terrestriality, and object manipulation; basically this hypothesis posits that manual lateralization was triggered by visually guided foraging for insects or mobile preys accompanied by a specialization of the right hand for postural control in ancestral primates. As a wide range of feeding strategies evolved, and changes in postural behavior occurred, the demands for controlling finer motor patterns (e.g., manipulative/extractive foraging) increased and the

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right hand gradually became the preferred hand for performing more elaborate and complex tasks (MacNeilage, 2007; MacNeilage et al., 1987). These hand preferences reflect specialization of the right and left hemisphere; the left hemisphere presides over sequential and planned motor actions, while the right is dominant for detecting and responding to unexpected movements (MacNeilage, 2007; Rogers et al., 2013). This hypothesis predicts that primate species with more complex manipulative skills and higher terrestrially will show a right-hand bias. Indeed, Meguerditchian et al. (2013) show that many terrestrial primate species display a right-hand bias for a coordinated bimanual task (the tube task), while more arboreal species tend to be left handed for the same task.

There is, however, a dearth of comparative studies among closely-related species and this type of data could provide a strong explanatory framework for between-species similarities and differences in hand preferences and could also be advantageous in identifying the factors that have selected for the extreme lateralization in hand use observed in humans (Corballis, 2017; Hopkins et al., 2015).

In this sense, capuchin monkeys could be excellent models to evaluate possible differences between species and the links with their ecology and foraging behavior. Alfaro, Silva Junior, & Rylands (2012) and Alfaro, Boubli, et al. (2012) after reviewing data on morphological, ecological, genetic, and behavioral evidence suggested the formal division of capuchins between the robust (tufted) capuchins, *Sapajus*, and the gracile (untufted) capuchins, *Cebus*. While the gracile and robust capuchin monkeys share a similar social structure, they show differences in the specifics of their diet. The robust capuchins are more specialized for destructive and extractive foraging, with a suite of morphological and behavioral adaptations associated with extractive foraging, while the gracile capuchins apparently are less dependent on extractive foraging and more dependent on patchy fruit resources (Alfaro, Silva Junior, & Rylands 2012; Janson & Boinski, 1992; Terborgh, 1983).

These capuchins provide a unique opportunity to study possible differences in hand preference during the tube task and to infer possible links with the habitat/ecology where they evolved. In this study, we investigated hand preference in seven species of capuchins (*Cebus albifrons*, *Cebus olivaceus*, *Sapajus apella*, *Sapajus flavius*, *Sapajus libidinosus*, *Sapajus robustus*, and *Sapajus xanthosternus*) in captivity. Furthermore, we investigated possible differences in extractive foraging among the robust and gracile capuchin using bamboos filled with sunflower seeds. Given the unique morphological adaptations of robust capuchins for extractive foraging, we predict they will show higher frequencies of extractive foraging compared to the gracile capuchins.

Westergaard & Suomi (1996a) hypothesized that capuchiṅś greater strength of hand preference, when compared to macaques, could be linked to their extractive foraging habits and associated complexity of manipulative activities when processing food (Fagot & Vauclair, 1991). Thus, we expect the gracile capuchins to show lower strength of lateralization when compared to the robust capuchins and that robust capuchins will also show a right-hand bias since extractive foraging and increased manipulative complexity is believed to favor the appearance of right hand preference (MacNeilage, 2007).

We hypothesize that *S. libidinosus* will show right hand preference and higher frequencies of extractive foraging than other capuchin species, since this is a species that evolved in the dry forests of Brazil (Cerrado & Caatinga, Alfaro, Boubli, et al., 2012). The exploitation of these drier habitats, with a mosaic distribution of food resources and a relatively unpredictable pattern of fruit availability, seems to exert a higher selective pressure for the use of feeding tools (Lee & Moura, 2015; Moura & Lee, 2004), for increased extractive foraging (Melin, Young, Mosdossy, & Fedigan, 2014; Moura, 2004) and seemed to have selected for increased use of terrestrial substrates (Wright et al., 2015); these factors are believed to favor right handedness (MacNeilage, 2007; Meguerditchian et al., 2013).

Within the robust capuchins we studied two species, *S. flavius* and *S. robustus*, endemics to the Atlantic forest (Silva, 2010; Vilanova, Silva-Junior, Grelle, & Cerqueira, 2005) and one species, *S. apella*, with a distribution range in the Amazonian rainforest. We expect these species to be more similar in their hand preference and extractive foraging since they evolved in rainforest habitat (Alfaro, Boubli, et al., 2012). The yellow breasted capuchin is usually considered to be restricted to the Atlantic forest, but it also occurs in dry forests (Canale, Guidorizzi, Kierulff, & Gatto, 2009). Individuals in captivity settings probably originated from populations with different selective pressures that might affect hand preference, for example, only populations from dry forest use tools (Canale et al., 2009). Thus, we expect hand preference and extractive foraging of yellow breasted capuchins (*S. xanthosternus*) to be in-between that of robust capuchins, which evolved in rainforest, and that of *S. libidinosus*.

2 | METHODS

2.1 | Subjects

Observations were made on 220 capuchin monkeys from seven species (Table 1). These capuchins were housed at zoological gardens, research facilities and "rescue" centers located in different regions of Brazil (Table 1). Enclosures in zoological gardens usually had environmental enrichment (supplementary material Figure S1), but two zoological gardens, Zoo Arruda Camara (Bica), and Zoo Dois Irmaos, also had capuchin monkeys kept in enclosures out of public view (quarantine) with poor environmental enrichment, consisting basically of one to two tree trunks on the floor. All individuals in the "quarantine" enclosures showed stereotypical behaviors (head twirl, pacing, and repetitive movements). In the Centro de Primatologia do Rio de Janeiro (CPRJ) monkeys were kept for conservationist purposes and for non-invasive behavioral studies. Although the enclosures had environmental enrichments and were located within an Atlantic forest remnant, some monkeys showed repetitive movements. In the RioZoo all the enclosures had small trees and toys, but some individuals showed stereotypical behaviors. The monkeys housed at the wildlife rescue center—CETAS from Paraiba, run by the Brazilian Institute of the Environment (IBAMA/ICMBio)—had enclosures located within a fragment of forest, but these had extremely poor (or were lacking) environmental enrichments and all the individuals showed

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TABLE 1 Total number of individuals for each species and facilities in which those species were hosted

Species	# individuals (infants)	# individuals that participated in the tube task	# individuals that participated in extractive foraging task	Facility
S. xanthosternos	53 (2)	37	51	CPRJ, ZooRio, Zoo Salvador, CENP
S. flavius	30 (2)	26	28	Zoo Arrruda Camera, Dois Irmaos, ZooRio, Zoo Salvador
S. libidinosus	42 (1)	35	41	Zoo Arrruda Camera, Dois Irmaos, CETAS
S. robustus	10 (1)	7	9	CPRJ, ZooRio
S. apella	61	33	61	CENP, Sauim Castanheira, CIGS, Zoo Tropical
Cebus albifrons	16 (2)	7	14	Rio Zoo, CENP, Sauim Castanheira
C. olivaceus	8	7	8	CENP
Total	220	152	212	

stereotypical behaviors indicative of stress. In the CETAS all the adult males but the alphas were castrated. The monkeys housed at the Centro Nacional de Primatas (CENP) in Belem were those with the worst wellbeing condition. All their enclosures were relatively smaller (3.6 × 2.2 × 2.3 m) than those elsewhere and they were located inside a large "barn," where the humidity and temperature were higher than the exterior causing a sensation of discomfort. These monkeys lacked environmental enrichment and were used in different types of research (invasive and non-invasive). The monkeys in CENP showed more stereotypical behaviors, were more aggressive and fearful than those housed elsewhere.

2.2 | Data collection

2.2.1 | Extractive foraging

For evaluating possible differences in extractive foraging across species we cut internodes of the giant bamboo (*Dendrocalamus giganteus*) measuring 35–40 cm and with diameter ranging from 6 to 10 cm. The internodes (bamboos hereafter) were filled with sunflower seeds. Each bamboo had about 10 randomly drilled small holes that permitted removal of some seeds (Supplementary material Figure S2). The bamboos stimulated the interest and curiosity of the capuchins and prompted the appearance of manipulative and extractive behaviors. The time lapse between the cutting of the bamboos and the handing of them over to the capuchins ranged from 1 week to 2 months.

The bamboos were placed within the enclosures in the afternoon (between 13 and 17 hr). The number of bamboos for each enclosure matched the capuchin's group size in order to avoid possible conflicts.

Allana collected the data using scan sampling and recorded all occurrences (Martin & Bateson, 1986) of foraging behavior directed toward the bamboos. Each observation session lasted 16 min and the behaviors were recorded at intervals of 2 min. The sessions were carried out for 3 days and each group had one session per day. Usually the bamboos were removed in the next day (early in the morning), but some facilities (Rio Zoo and CENP) asked the bamboos to be removed on the same day. We recorded the following extractive foraging behaviors: banging the bamboo against the substrate, bitten, ripped apart using the teeth, torn apart using the hands. We also recorded the state of the bamboos retrieved in the same or on the next day since it could give clues on the "modus foraging" of each species. We categorized these bamboos as: (i) broken bamboos—those lacking pieces and open, or partially open, with an exposed interior; (ii) cracked bamboos—those splintered and/or fissured; and (iii) lacerated bamboos—those with the fibrous exterior shredded, but without exposing the interior.

2.2.2 | Hand preference

We assessed hand preferences using the tube task (Hopkins, 1995). Peanut butter mixed with smashed bananas were smeared on the insides of polyvinyl chloride (PVC) tubes measuring 9 cm in length and 22 mm in diameter. The reward (peanut butter + banana) was smeared near the middle of the tube, which compelled the subjects to use one hand to hold the tube and the other hand to remove the mixture. The tubes were linked by a chain (about 1 m) to a pipe (1 m length) placed outside the enclosure on the floor (Supplementary material Figure S3), to prevent the monkeys carrying-away the tubes and to retrieve them easily after an experimental session. The number of tubes introduced ranged from 6 to 9 according to the group size in the enclosures, which also prevented monopolization of the tubes by the dominant individuals. For all the tests, the tubes were placed on the ground of the enclosures/cages simultaneously. All individuals were tested in their social groups, and only four individuals were housed individually.

Data on each monkey's hand preference were collected by Allana using focal animal sampling with a continuous record of hand-use in 3 min sessions (Martin & Bateson, 1986). If the individuals could not get more food from the tube the session was stopped briefly and the tubes re-filled. All individuals received one session of the tube task per day and were observed for three consecutive days. The hand of the finger used to extract the mixture from the tube was recorded either as right or left using both frequencies (i.e., # times the finger of one given hand was inserted into the tube and brought into the mouth) and bouts (a continuous sequence of tube manipulations). We defined a bout of hand-use as the sequence of identical actions (i.e., when the finger was inserted into the tube) and recorded only the first event of the sequence. A new bout occurred only if the individual dropped the tube or changed the hand holding the tube (Maille et al., 2013; Molesti et al., 2016).

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2.3 | Data analysis

We calculated a handedness index (HI) from the frequency data and from bouts (HI bouts) for each individual by using the following formula: (#R−#L)/(#R + #L). We included only individuals with frequencies ≥6 and with at least six bouts. The HI produces scores ranging from −1 to +1. A right-hand bias is reflected by a positive HI, while negative scores indicate a preference for the left hand. We used absolute values of HI to characterize strength of lateralization, independent of direction.

To characterize group level hand preference a one sample *t*-test was applied to HI scores on each species (Hopkins, 1999, 2013). We followed Hopkins (2013) and used cut-points, in a non-continuous approach, of HI ≤−0.2 to characterize left-handed individuals and HI ≥0.2 to identify right-handed individuals, while the remaining individuals were classified as ambi-preferent; then using a chi square goodness of fit test (Hopkins, 1999) we determined if there was significant hand preference. We used a between-subjects *t*-test to examine the effects of individual age and sex on hand preference, and for comparison between *Cebus* and *Sapajus*. To verify differences among the groups' means we used ANOVA.

We followed the recommendations of Field (2009) for analysing the data and we used SPSS v 13 to carry out all the analyses. All tests were two-tailed, and the significance level set at $p = 0.05$. For reporting effect size (Cohen's *d*) we followed Lakens (2013) and we used the following formula for one sample *t*-test: $d_z = \frac{t}{\sqrt{n}}$ and for between subjects (independent *t* test):

$$d_s = \frac{\bar{x}1 - \bar{x}2}{\sqrt{\frac{(n1-1)SD_1^2+(n2-1)SD_2^2}{n1+n2-2}}}$$

Data from all the capuchins were obtained only after the study purpose and design were approved by the administrative and ethics committees at the different institutions. The type of experiment we carried out improves the well-being of the monkeys, since it stimulates their interest and curiosity and promotes the appearance of manipulative and extractive behaviors. This research adhered to the American Society of Primatologists' principles for the ethical treatment of primates.

3 | RESULTS

3.1 | Extractive foraging

The capuchin monkeys showed marked differences ($F_{(6,\ 205)} = 10.82$, $p < 0.0001$) in relation to extractive foraging. Two species, *S. flavius* and *S. robustus*, used extractive foraging most prominently (Figure 1). They showed significantly higher frequencies of extractive foraging than *S. apella*, *S. libidinosus*, and *C. albifrons* (Table 2).

As we expected species that evolved in similar habitats to not differ, we pooled their data by habitat and re-run the analysis. All *Sapajus* from Atlantic forest showed higher frequencies of extractive foraging than *S. apella* and *Cebus* spp (Figure 2).

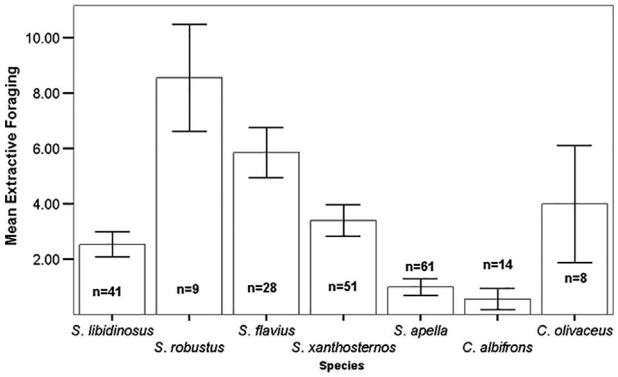


FIGURE 1 Species differences in the mean frequency of extractive foraging. Bars show mean and error bar show ±1 SE. Individuals that did not manipulate the bamboo were included in the analyses as well

S. robustus and *S. flavius* had proportionally more damaged bamboos in their cages after the experiment than the other species (Table 3). It is noteworthy that *S. apella* were more similar to *Cebus spp* than other *Sapajus*; no destroyed bamboos were found in their cages and most individuals preferred not to interact with the bamboos (Table 3). Overall, *Sapajus*, with the exception of *S. apella*, were more curious and approached bamboos more readily when compared to *Cebus*.

3.2 | Sex and age differences

When we analyzed differences in extractive foraging among age and sex groups, we found that males tended to engage in extractive foraging more often than females (Suplementary material Table S1), but the difference was significant only for *S. apella* ($t = 2.31$; df = 33.97; $p = 0.027$, equal variances not assumed) and *S. xanthosternus* ($t = 3.45$; df = 30.82; $p = 0.002$, equal variances not assumed). Likewise, adults showed higher frequencies of extractive foraging when compared to juveniles/subadults (Suplementary material Table S1), but the difference was significant only for *S. apella* ($t = 3.28$; df = 35.45; $p = 0.002$, equal variances not assumed), and *S. xanthosternus* ($t = 4.59$; df = 45.24; $p < 0.001$, equal variances not assumed).

TABLE 2 Species differences in frequency of extractive foraging.

Species	*p* value from Games Howell post hoc test
S. flavius > *S. apella*	0.000
S. flavius > *S. libidinosus*	0.035
S. flavius > *C. albifrons*	0.000
S. libidinosus > *C. albifrons*	0.03
S. robustus > *S. apella*	0.045
S. robustus > *C. albifrons*	0.033
S. xanthosternus > *S. apella*	0.007
S. xanthosternus > *C. albifrons*	0.002

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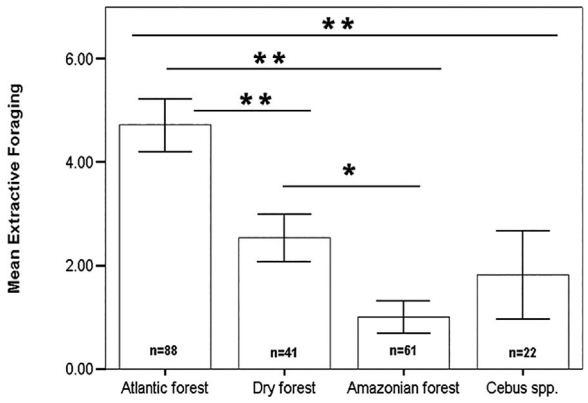


FIGURE 2 Extractive foraging by the forest habitat where the different capuchins' species evolved. Bars show mean and error bar show ±1 SE. Games-Howell post hoc test: * = 0.03, ** < 0.001

3.3 | Hand preference

The HI scores calculated based on frequencies compared to the HI scores calculated from bouts were almost identical, indicating that these two approaches give similar results (Figure 3). Further, most papers in which the tube task paradigm was used, particularly those on capuchin monkeys, reported HI scores based on frequencies (e.g., Lilak & Phillips, 2008; Meunier & Vauclair 2007; Westergaard & Suomi, 1996a). Thus, we considered only the HI scores based on frequencies for the analyses. Over 92% of the individuals had frequencies of hand-use higher than seven (mean = 31.4; range: 6–119).

There was a non-significant trend for the species to differ in the HI scores ($F_{(6,\ 145)} = 1.97$; $p = 0.072$). The abs-HI, which indicates the strength of laterality, did not differ across species ($F_{(6,\ 145} = 1.5$; $p = 0.18$). Since the capuchin species from the Atlantic forest (*S. flavius*, *S. robustus*, and *S. xanthosternus*) showed higher frequencies of extractive foraging than species from Amazonia and dry forest we pooled their data by habitat, and we also collapsed the data points of the two *Cebus* species for more robust comparisons and to increase statistical power. The type of forest where capuchins evolved seemed to affect hand preference ($F_{(3,\ 148)} = 2.88$; $p = 0.038$) and had an effect, albeit non-significant ($F_{(3,\ 148)} = 2.52$; $p = 0.06$), on strength of laterality. A Games–Howell post hoc test indicated that Atlantic forest capuchins had a significant right-hand preference in relation to the gracile capuchins ($p = 0.002$) and the gracile capuchins had a more pronounced leftward hand preference when compared to *S. apella* ($p = 0.056$).

To check for population level hand preference we performed one-sample *t*-tests on the HI scores within each species. In three species the HI scores differed from a chance distribution with a mean of zero: *C. albifrons* ($t = -2.63$; df = 6; $p = 0.039$; $d_z = 0.99$), *C. olivaceus* ($t = -4.51$; df = 6; $p = 0.004$; $d_z = 1.7$), and *S. libidinosus* ($t = -2.3$; df = 34; $p = 0.027$; $d_z = 0.39$).

Using the cut-points for hand preference we classified the individuals as right-handed, left-handed, and ambi-preferent and we compared the number of left and right-handed monkeys within each species (Table 4). Within the *S. libidinosus* sample, the chi-square goodness-of-fit tests revealed significant differences in the number of right- and left-handed individuals ($\chi^2 = 7.5$; df = 1 $p < 0.01$). For the other capuchin monkey species (only those with at least five individuals in each category), there was no difference between the number of left and right-handed individuals.

3.4 | Sex and age differences

S. apella was the only species to show sex differences in the HI scores (Figure 4). Females had higher HI scores than males for using the right hand ($t = -2.24$; df = 22.94; $p = 0.035$, equal variances not assumed, $d_s = 0.32$). We did not find sex differences in the HI scores and strength of laterality in the remaining species (*t* test, $p > 0.2$, Suplementary material Table S2). The capuchin species tested did not show age differences in the HI and ABS HI scores (Suplementary material Table S2).

Sapajus and *Cebus* differed in their laterality, *Cebus* showed a more pronounced leftward bias than *Sapajus* ($t = -3.71$; df = 24.38; $p = 0.001$, equal variance not assumed, $d_s = 0.62$). They did not differ in the strength of laterality ($t = -1.43$; df = 150; $p = 0.15$).

TABLE 3 Proportion of damaged bamboos found in the precincts

Bamboos' removal	Species	# bamboos	% broken bamboos	% cracked bamboos	% lacerated bamboos	% undamaged
Next day	*S. xanthosternos*	93	16.1	3.2	9.7	71
	S. flavius	69	10.1	1.4	2.9	85.6
	S. libidinosus	129	14.7	1.6	3.1	80.6
	S. robustus	15	13.3	6.7	13.3	66.7
	S. apella	51	0	0	0	100
Same day	*S. xanthosternos*	66	1.5	1.5	4.5	92.5
	S. flavius	21	0	4.8	33.3	61.9
	S. robustus	15	0	6.7	0	93.3
	S. apella	132	0	0	0	100
	Cebus spp.	81	0	0	0	100

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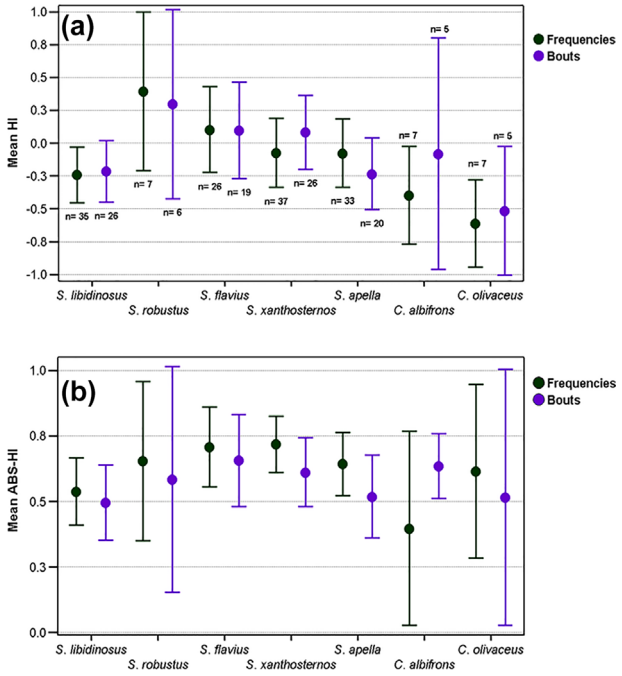


FIGURE 3 Comparison of average HI scores (a) and average ABS-HI scores (b) based on frequencies and bouts. Dots show mean and error bar show 95%CI

4 | DISCUSSION

Some of the capuchin monkey species studied showed a rightward bias in the mean HI scores, while others were left biased, but the difference was non-significant. Capuchins from the Atlantic forest showed a significant right-hand preference in relation to the *Cebus* spp. There was no significant difference in the strength of laterality across species. In the extractive foraging, however, there was a clear difference; *S. robustus* and *S. flavius* surpassed the other capuchins and the capuchins from the Amazonian forest, whereas *S. libidinosus* tended to show low frequencies of extractive foraging. Males and adult individuals tended to be the most prolific extractive foragers, which is in agreement with findings in wild groups (e.g., Gunst, Boinski, & Fragaszy, 2008; Rose, 1994; Terborgh, 1983). We did not find age differences either in strength of laterality or in the direction of laterality, which agrees with similar studies (Spinozzi et al., 1998; Westergaard & Suomi, 1996a). Although Spinozzi et al. (1998) found a significant group-level right-hand bias in brown capuchins, other authors did not report hand preference in the tube task (Lilak & Phillips, 2008; Westergaard & Suomi, 1996a). This inconsistency may reflect specifics of the tube task used by these authors. Spinozzi et al. (1998) used transparent tubes that were hanging by a chain from the cage ceiling, which could have increased the complexity of the task and may have favored a group level right-hand bias (see Schweitzer et al., 2007).

Interestingly, *S. apella* was the only species to show sex differences in laterality, with females favoring their right hand and males preferring their left hand. Spinozzi et al. (1998) also reported that females are significantly more right handed than males when tested with the tube task. This apparent sex difference in brown capuchin monkeys has also been noted in a tool using context; in a nut cracking task the males tend to be left handed and females show a strong right-hand preference (Westergaard & Suomi, 1993; but see Westergaard & Suomi, 1996a). Likewise, in a task involving prey capture, male brown capuchins showed a significant left hand preference (Hellner-Burris, Sobieski, Gilbert, & Phillips, 2010). The reasons underlying the sex differences in *S. apella* in relation to hand preference are not clear, however. It is possible that selective pressures linked to sex differences in foraging could account for these results (Hellner-Burris et al., 2010; Melin, Fedigan, Young, & Kawamura, 2010; Pouydebat, Borel, Chotard & Fragaszy, 2014; Rose, 1994). Nevertheless, the artificial environments of captivity with less variation and diversity in substrate could be a source of stress (Morgan & Tromborg, 2007). Some authors report that glucocorticoid hormones, related to the body's response to stress, could modulate behavioral lateralization and seem to impact on hand preference (Bennet, Suomi, & Hopkins, 2008; Ocklenburg, Korte, Peterburs, Wolf, & Güntürküna, 2016; Rogers et al., 2013; Westergaard, Byrne, & Suomi, 2000). There are possibly sex differences in stress reactivity in *S. apella*, during their development, that may affect the levels of glucocorticoid hormones and influence the emergence of hand preference (Ocklenburg et al., 2016). Unfortunately, we lack data on plasma cortisol concentration and detailed information on the origins and rearing conditions of the monkeys in most facilities that preclude more detailed analysis. Thus, any potential explanation for the sex differences we found in hand preference is only speculative, but clearly warrants further investigation.

TABLE 4 Hand preference based on the HI cut-points

Species	Left	Ambidextrous	Right	Total
S. libidinosus	20	9	6	35
S. robustus	1	1	5	7
S. flavius	9	5	12	26
S. xanthosternos	18	5	14	37
S. apella	16	5	12	33
C. albifrons	4	3	0	7
C. olivaceus	6	1	0	7

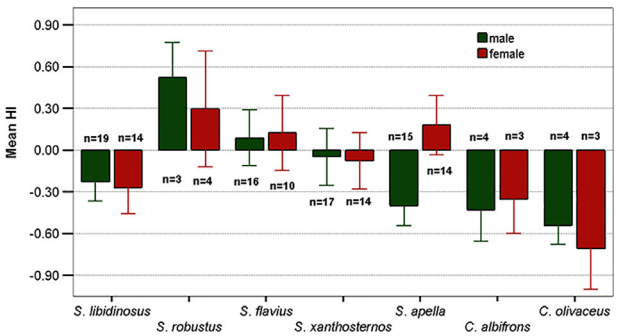


FIGURE 4 Comparison of sex differences in hand preference. Bars show mean and error bar show ±1 SE. Note that the total # individuals for some species is different from Figure 3 because some individuals could not be sexed during observations and lacked records in the facilities

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4.1 | Extractive foraging, hand bias, tool use, and the weirdness of *Sapajus libidinosus*

Extractive foraging is an activity that involves complex manipulation and often requires the use of sequential specific actions (e.g., Cochet & Byrne, 2013; Gunst et al., 2008; Parker & Gibson 1977) and many papers indicate it could be a driver for increased laterality and handedness (Brown & Magat, 2011; Fagot & Vauclair, 1991; Heldstab et al., 2016; MacNeilage, 2007). Moreover, the manipulative complexity of extractive foraging seems to intensify during tool use, which requires diverse manual actions combined with strength, precision, bimanually coordinated movements, and more cognitive skills (Heldstab et al., 2016; Parker & Gibson, 1977; Resende, Nagy-Reis, Lacerda, Pagnnota & Savalli, 2014). Tool use is considered as a driver for right handedness (e.g., Forrester, Quaresmini, Leavens, Mareschal, & Thomas, 2013; Frost, 1980; Hopkins et al., 2017; Moura, 2015). Thus, we expected that *S. libidinosus*, a species that evolved in the dry forests of Brazil (Cerrado & Caatinga, Alfaro, Boubli, et al., 2012) should be more right handed. Yet, these capuchin monkeys showed a left-hand bias at the population level and were not the most prominent extractive foragers.

Our results for *S. libidinosus* could be due to the small sample size and captive condition, but we cannot rule out that the preponderance for left hand use could also be an evolutionary by-product of the unique milieu of Caatinga/Cerrado dry forest that gave rise to this capuchin species. Morphological data show that *S. libidinosus* is the most distinct among all robust capuchins, probably as a consequence of the unique selective factors in the Caatinga/Cerrado dry forest environment that has selected for greater reliance on terrestriality and the use of tools (Moura & Lee, 2004; Wright et al., 2015). Our data, however, suggest that tool use might not be a key driver for hand preference in *Sapajus* as previously suggested (Moura, 2015; Westergaard & Suomi, 1996b).

On the other hand, two capuchin species from the Atlantic forest, had positive mean values of HI and showed significantly higher frequencies of extractive foraging when compared to species from dry forest and Amazonia wet forest. These are puzzling results. We were not expecting positive HI values nor a higher frequency of extractive foraging than those we found for the Atlantic forest capuchins. We cannot rule out the possibility of these results being an artefact of captive settings and small sample size. Nevertheless, there are reports of wild capuchins in the Atlantic forest showing a right-hand preference in a foraging context. For example, Souto et al. (2011) report three individuals of *S. flavius* using sticks during foraging and they always hold the stick in their right hand. Likewise, individuals in a wild group of *S. nigritus*, a closely related species to S. *robustus*, showed a significant preference for using the right hand in a manipulative task (Garber, Gomes, & Bicca-Marques, 2008).

Capuchins from the Atlantic forest seem to rely heavily on plant material, particularly the leaf base of bromeliads, and palm petioles (Brocardo, Gonçalves, Zipparro, & Galetti, 2010; Izar et al., 2012) that may require specific motor demands associated with the complementary use of both hands. These populations of capuchins spend significantly more time consuming leaves than capuchins from the Caatinga dry forest, which tend to spend more time foraging for mobile prey (Izar et al., 2012). It is possible that capuchin monkeys from the Atlantic forest evolved under a stronger selective pressure for high reliance on bimanual feeding and extractive foraging, which requires more sequential actions and planning and might put a stronger selective pressure on left hemisphere/right hand use (Cochet & Byrne, 2013; Hopkins et al., 2017). Unfortunately, there is no published study on the feeding ecology of *S. flavius* and detailed data on manipulative and extractive foraging techniques employed by Atlantic forest capuchins is also lacking. These differences in diet might also be the ultimate explanation for the preponderant use of the left hand by *S. libidinosus*. This species probably evolved with a greater reliance on fast mobile prey such as lizards, which are particularly abundant and available year-round in dry environments (Morton & James, 1988). In this scenario, group-level left-hand bias might be expected since capturing a fast-moving prey could depend on the integration of the speed and trajectory of the prey with a quick forelimb movement, which is under control of the right hemisphere (Hellner-Burris et al., 2010; MacNeilage, 2007; Pouydebat et al., 2014). Although our reasoning is speculative, it deserves further investigation.

We found significant differences at the genus level. The gracile capuchins showed significant higher left-hand bias and showed lower frequencies of extractive foraging when compared to the robust capuchins, although they were comparable to *S. apella*. All the capuchins species are highly arboreal and according to the postural origin theory (MacNeilage et al., 1987), they were expected to show a similar left-hand preference. Yet they did differ and the task complexity hypothesis (Fagot & Vauclair, 1991; MacNeilage et al., 1987) provides the most reasonable explanation for the observed differences. The gracile and robust capuchins differ markedly in their feeding ecology, with gracile capuchins being more dependent on fruits, while the robust capuchins rely more on extractive foraging and durophagy (Alfaro, Boubli, et al., 2012; Janson & Boinski, 1992; Terborgh, 1983; Wright et al., 2015). Our data on extractive foraging suggests *S. apella* seemed to converge with the gracile capuchins, probably due to evolution under similar ecological pressures.

While our analysis indicates a weak significant impact of species on hand preference, caution is required when interpreting these results since we cannot rule out some unforeseen effects of captive settings. Additional insights into the ecological forces driving hand preference differentiation among capuchin species can be gained through collection of more detailed behavioral and ecological data for each of the species. Unfortunately, there are just four studies evaluating hand preference in wild capuchins (Garber et al., 2008; Moura, 2015; Panger, 1998; Panger & Wolfe, 2000) and further studies in wild populations, particularly those from the Atlantic forest, are clearly needed. Our dataset on hand preference (Supplementary material Table S3) and further additional data could provide a stronger empirical framework for testing broad hypothesis on the selective pressures shaping the evolution of handedness.

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SUPPORTING INFORMATION

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